

Representation, Topology, and a Blood–Brain-Barrier-Associated Slow-Wave Marker: A Controlled Graph Neural Network Study of EEG in Alzheimer’s Disease and Frontotemporal Dementia

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Abstract

Objective. EEG-based classification of Alzheimer’s disease (AD) and frontotemporal dementia (FTD) on the same public dataset reports accuracies from 80% to 99%, but studies change the signal representation, graph construction, model and evaluation protocol at once. We isolated these factors and tested a blood–brain-barrier (BBB)-associated EEG marker. *Approach.* Using OpenNeuro ds004504 (36 AD, 23 FTD, 29 controls), we crossed seven node-feature representations with three graph constructions under one fixed two-layer graph convolutional network, evaluated on 25 frozen, paired subject-level cross-validation splits (525 runs), with non-graph baselines, 1,700 ablation and permutation runs, and a leakage experiment. We also characterised paroxysmal slow-wave events (PSWEs), an EEG marker linked to BBB dysfunction. *Main results.* Representation had a large effect on three-class macro-F1 (partial $\eta^2 = 0.33$); graph construction had a small one ($\eta^2 = 0.14$) and no interaction. The best configuration (wavelet features, spatial graph) reached 0.586 (95% CI 0.538–0.628; permutation $p = 0.005$), matching but not exceeding non-graph classifiers. Window-level splitting of the identical pipeline raised it to 0.988. Fixed-threshold PSWEs were more frequent in AD (rate ratio 2.96) and FTD (2.07) than in controls, but this was explained by background slowing ($\rho = 0.88$); relative to each person’s background, transient slowing events were less frequent in dementia. *Significance.* On this benchmark the signal representation and, above all, the evaluation protocol — not the graph or the model — determine reported performance, and an EEG marker of BBB dysfunction must be defined relative to background slowing. Code, frozen folds and all raw runs are released.

Keywords: EEG; graph neural network; Alzheimer’s disease; frontotemporal dementia; signal representation; functional connectivity; blood–brain barrier; paroxysmal slow-wave events; data leakage; reproducibility

1 Introduction

Alzheimer’s disease (AD) and frontotemporal dementia (FTD) are leading causes of neurodegenerative dementia, particularly with onset before 65, and telling them apart is clinically consequential: they differ in prognosis, in care needs and in which disease-modifying treatments are appropriate. Scalp electroencephalography (EEG) is attractive for this task because it is cheap, non-invasive and available in routine care, in contrast to amyloid PET or MRI. Its physiological basis is well established: AD is increasingly understood as a disorder of large-scale brain networks [1], and graph-theoretical EEG

studies have shown that AD and FTD disturb those networks in *different* ways [2], which is what makes a three-class AD/FTD/control problem scientifically meaningful rather than merely harder.

Machine-learning work on this question has moved from spectral markers and handcrafted features with classical classifiers [3] to convolutional networks and transformers [4, 5] and, most recently, to graph neural networks (GNNs) that treat electrodes as nodes and inter-electrode coupling as edges [6–12]. The public OpenNeuro dataset ds004504 [13, 14], with 88 subjects recorded on one device under one protocol, has become the de-facto benchmark. Reported accuracies on this single dataset span roughly 80–99%.

That spread is implausibly wide for identical data, and it hides a methodological problem. Published studies change several things at once: (i) the *signal representation* fed to the model, (ii) the *graph construction* that defines which electrodes are connected, (iii) the *model architecture*, and (iv) the *evaluation protocol*. Independent evidence suggests that the last of these matters most. A 2026 scoping review of this exact benchmark concluded that apparent deep-learning gains “arise not from inherent representational advantage, but from data leakage” [15], echoing a broader warning about leakage across machine-learning-based science [16]. A 2026 comparison of five time–frequency representations under a single convolutional network [17] isolated representation, but its authors note that it “does not explicitly model the true spatial topology” and that graph-based schemes remain unexplored; others observe that most studies “overlook spatial connectivity between channels” [10], and explicitly call for graph convolutional networks in future work [18, 19]. No study has varied representation and graph construction together while holding the model and the evaluation protocol fixed.

A second, biologically motivated gap concerns the blood–brain barrier (BBB). BBB breakdown is an early biomarker of cognitive dysfunction [20, 21], and Milikovsky et al. described *paroxysmal slow-wave events* (PSWEs) — episodes in which the EEG median power frequency falls below 6 Hz for at least 5 s — in patients with AD, linked them to cognitive impairment, localised them to cortex with BBB dysfunction in epilepsy, and induced them in rodents by albumin extravasation [22, 23]. PSWEs have since been examined in Parkinson’s disease [24] and epilepsy [25], but, to our knowledge, never in FTD and never on an open dementia cohort. Whether this BBB-associated EEG marker carries diagnostic information beyond ordinary EEG slowing has not been tested.

We therefore ran a controlled factorial experiment. Seven node-feature representations (spectral, connectivity, wavelet, time-domain, raw, fused, and PSWE-derived) were crossed with three graph constructions (spatial, functional, hybrid) under a single fixed two-layer graph convolutional network [26], with subject-level repeated cross-validation, identical frozen folds for every condition, and hyperparameters fixed before any result was seen. Because the model is the control variable rather than the contribution, it is deliberately small. Around this core we measured the inflation produced by window-level (leaky) splitting on the identical pipeline, quantified age/sex confounding, and characterised PSWE burden across AD, FTD and controls.

Our contributions are:

1. **Methodological.** A fully crossed 7×3 protocol that separates signal representation, graph topology and model architecture, with paired statistics over 25 frozen subject-level test splits and ablations that test whether the graph matters at all.
2. **Empirical.** Effect sizes and confidence intervals for which representations a GNN benefits from in three-class AD/FTD/control classification, with non-graph twins and a demographic confound floor as reference points.
3. **Protocol.** A direct measurement, on one pipeline, of how much window-level splitting inflates results — a calibration factor for reading published numbers on this benchmark.

4. **Biological.** The first characterisation of BBB-associated PSWEs in FTD, an independent-cohort test of the AD finding, and a test of whether the marker is separable from background slowing.
5. **Reproducibility.** Code, seeds, frozen folds and every raw run released; the full study re-runs on free cloud CPUs.

2 Related work

Signal representations for EEG dementia classification. Spectral representations dominate: relative band power and EEG slowing are the most replicated AD markers [4, 27, 28]. Time-domain and complexity features (Hjorth parameters, entropies) are cheap and interpretable [5, 28]. Time-frequency transforms such as the continuous or discrete wavelet transform retain non-stationary structure that a power spectrum averages away [17, 29], and several recent studies encode EEG as images for convolutional networks [11, 17, 30]. Representation choice is thus treated as a first-class question — but almost always with convolutional models, and without varying the spatial structure the model sees.

Graph neural networks for EEG. Spectral graph convolution [26] and attention-based variants [31] underpin most EEG GNNs, reviewed in [6]. For dementia, GNNs have been built on functional-connectivity graphs [7], learned adjacency [8], multi-connectivity inputs [12], time-resolved graphs [11] and spatial-functional hybrids [10]; for Parkinson’s disease, interpretable graph learning has been used to localise abnormal connectivity [19]. The closest precedent [7] compared connectivity measures as GNN inputs but held node features fixed. A 2025 preprint applied a graph transformer to ds004504 for FTD versus controls with window-level prediction [32]. None of these studies crosses node-feature representation with edge definition under a fixed model.

Graph construction for EEG. Edges may be defined spatially (inter-electrode distance), functionally (statistical coupling), by learning, or by combinations of these. Phase-lag-based measures [33, 34] suppress the zero-lag coupling produced by volume conduction, which otherwise inflates functional graphs; weighted graphs are recommended over binarised ones for EEG [2], and phase-synchronisation networks differ between dementia subtypes [29, 35]. Spatial graphs carry no subject-specific information on their own and act as a smoothing prior; whether that prior helps is an empirical question we test directly.

Evaluation protocol and leakage. Leakage — information from test data reaching training — is a general failure mode of machine-learning-based science [16]. In EEG it takes a specific form: windows cut from one recording are highly correlated, so splitting *windows* rather than *subjects* lets a model recognise individuals instead of disease. The 2026 scoping review of ds004504 attributes the upper end of reported accuracies to this effect [15]. We measure it directly on one pipeline.

EEG and the blood–brain barrier. BBB breakdown precedes and predicts cognitive decline [20, 21]. Milikovsky et al. proposed PSWEs as an EEG manifestation of BBB-related cortical dysfunction in AD and epilepsy [22], with a companion study tracing the mechanism to albumin-activated TGF β signalling [23]. Subsequent work found PSWEs uncommon in Parkinson’s disease [24] and explored them as an epilepsy biomarker [25]. Because ds004504 includes no BBB imaging, we treat PSWEs strictly as a *BBB-dysfunction-associated* EEG marker, not as a measurement of BBB integrity.

Table 1: Positioning against the closest prior work on EEG dementia classification. ✓ = yes; – = no; ? = not yet verified against the full text.

Study	Varies repr.	Varies graph	Fixed model	Subject-level split	Public data
DICE-Net [4]	–	–	✓	✓	✓
FC-measure GNN [7]	–	✓	✓	?	?
Adaptive GCN [8]	–	✓	–	?	?
TFR comparison [17]	✓	–	✓	?	✓
Spatial-functional GNN [10]	–	–	–	?	?
Multi-connectivity GCN [12]	–	✓	–	?	?
Graph transformer [32]	–	–	✓	✓	✓
This study	✓	✓	✓	✓	✓

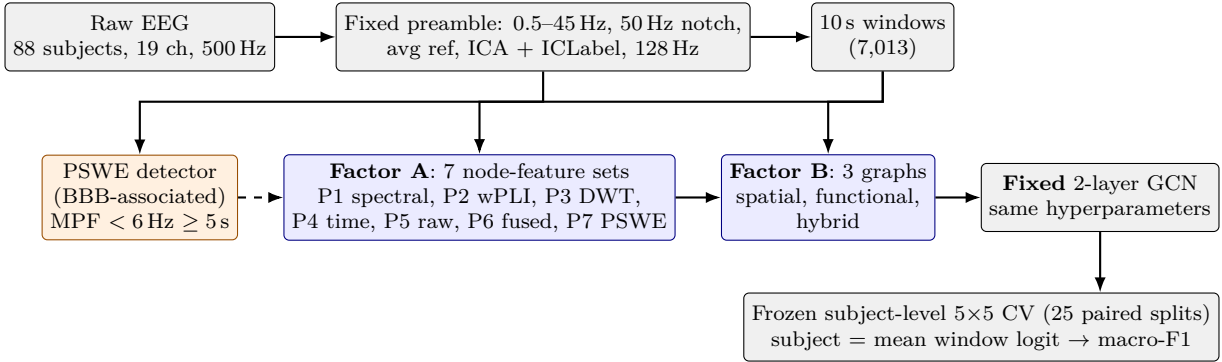


Figure 1: Study design. Grey components are held fixed; only the node features (Factor A) and the graph construction (Factor B) vary, in a fully crossed 7×3 design evaluated on the same 25 frozen subject-level splits. The PSWE detector (orange) supplies the blood–brain–barrier-associated marker, used as P7 node features and, in an ablation, as a co-occurrence graph.

Positioning. Table 1 summarises how this study differs from the closest prior work.

3 Materials and methods

Figure 1 summarises the design.

3.1 Dataset

We used OpenNeuro ds004504 [13, 14]: resting-state, eyes-closed scalp EEG from 88 participants recorded at AHEPA University Hospital (36 AD, 23 FTD, 29 cognitively normal controls, CN). All values below were read from the dataset’s own BIDS files rather than from the descriptor paper. Recordings use 19 electrodes of the 10–20 system (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz) referenced to A1–A2, sampled at 500 Hz on a Nihon Kohden EEG 2100 system, with a 0.4–50 Hz acquisition filter. Recording length per subject ranges from 307 to 1291 s (mean 802 s; 19.6 h in total). Group demographics are given in Table 2. The data are released under CC0; the study uses only de-identified public data. The dataset’s own metadata cite a version-1.0.9 DOI that did not resolve when checked; we cite version 1.0.8, whose EEG data are identical (the 1.0.9 changelog records only a README edit).

Table 2: Participant characteristics (mean $\pm$ SD), from `participants.tsv`. MMSE is reported for description only and is never a model input.

	AD ($n = 36$)	FTD ($n = 23$)	CN ($n = 29$)
Age (years)	66.4 ± 7.9	63.7 ± 8.2	67.9 ± 5.4
Sex (F/M)	24/12	9/14	11/18
MMSE	17.8 ± 4.5	22.2 ± 2.6	30.0 ± 0.0

3.2 Fixed preprocessing

Every condition shares one code path, implemented with MNE-Python [36]. Raw recordings were band-pass filtered at 0.5–45 Hz, notch filtered at 50 Hz, and re-referenced to the common average. Independent component analysis (extended Infomax [37], 18 components — one fewer than the channel count because of the average reference) was fitted on a 1 Hz high-passed copy of each recording, and components that ICLabel [38] assigned to eye or muscle artefact with probability ≥ 0.8 were removed. Data were then resampled to 128 Hz and cut into non-overlapping 10 s windows (1,280 samples), giving 7,013 windows in total (30–129 per subject). ICA is fitted per recording, without labels or other subjects’ data, and therefore cannot leak information across cross-validation folds; every transform that is estimated across subjects (feature scaling, demographic residualisation) is fitted on training subjects only.

3.3 Factor A: node-feature representations

Each window is a graph whose 19 nodes are the electrodes. Seven representations define the node features $X \in \mathbb{R}^{19 \times F}$:

- P1, spectral** ($F = 7$). Relative power in five bands (delta 0.5–4, theta 4–8, alpha 8–13, beta 13–30, gamma 30–45 Hz) from Welch’s method [39] (2 s Hann segments, 50% overlap), normalised spectral entropy, and peak frequency within 4–15 Hz.
- P2, connectivity** ($F = 5$). Node strength of the weighted phase-lag index (wPLI) [33, 34] in each band: the mean wPLI between a channel and the other 18.
- P3, wavelet** ($F = 20$). Five-level Daubechies-4 discrete wavelet transform [40]; for the sub-bands A5, D5, D4, D3 and D2 (D1, 32–64 Hz, lies above the low-pass cut-off), relative energy, normalised coefficient entropy, mean absolute coefficient and standard deviation.
- P4, time domain** ($F = 10$). Mean, variance, root mean square, skewness, kurtosis, zero-crossing rate, Hjorth mobility and complexity [41], sample entropy [42] ($m = 2$, $r = 0.2\sigma$) and line length. Line length replaces Hjorth activity, which equals the variance.
- P5, raw** ($F = 1280$). The window z-scored per channel.
- P6, fused** ($F = 34$). P1, P3 and P2 concatenated, plus each node’s weighted clustering coefficient [43] and eigenvector centrality in the alpha-band wPLI graph.
- P7, PSWE** ($F = 4$). From the PSWE detector (Section 3.8): fraction of the window spent in a PSWE, mean and minimum median power frequency in the window, and the local PSWE onset rate within ± 60 s.

The wPLI between channels i and j was computed within each window from band-passed (4th-order Butterworth, zero-phase) analytic signals $z_i(t)$ as

$$\text{wPLI}_{ij} = \frac{|\langle \text{Im}(z_i z_j^*) \rangle_t|}{\langle |\text{Im}(z_i z_j^*)| \rangle_t}, \quad (1)$$

which is insensitive to zero-lag coupling from volume conduction [34].

3.4 Factor B: graph construction

Three edge definitions were crossed with every representation. *Spatial*: a Gaussian kernel $w_{ij} = \exp(-d_{ij}^2/2\sigma^2)$ on Euclidean distances between standard 10–20 electrode positions, with σ the median pairwise distance; this graph is identical for every window and subject. *Functional*: the alpha-band wPLI of the window. *Hybrid*: the element-wise product of the two. Each graph was sparsified by keeping the $k = 4$ strongest edges per node, symmetrised as $A = \max(A, A^\top)$, kept weighted, and normalised as $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$.

3.5 The fixed model

The model is the control variable, not the contribution, so a single small graph convolutional network (GCN) [26] was used everywhere: $H_1 = \text{Dropout}_{0.5}(\text{BN}(\text{ReLU}(\hat{A}XW_0)))$, $H_2 = \text{ReLU}(\hat{A}H_1W_1)$, mean pooling over the 19 nodes, and a linear layer to three classes, with $W_0 \in \mathbb{R}^{F \times 64}$ and $W_1 \in \mathbb{R}^{64 \times 32}$. This gives 2,819 parameters for $F = 7$ and 3,651 for $F = 20$; P5 is the unavoidable exception (84,355 parameters, because $F = 1280$). Two hops of propagation already reach most of a sparsified 19-node graph, and deeper GCNs risk over-smoothing; depth is tested as an ablation. Graph attention networks [31] learn edge weights and would partly undo the edge definitions being compared, so a GAT is used only as a sensitivity check.

Training used AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch size 64, and at most 200 epochs with early stopping (patience 20) on validation macro-F1 computed at subject level. The loss was weighted by inverse class frequency (over subjects) divided by each subject’s window count, so every subject contributes equally regardless of recording length. A subject’s prediction is the mean of its window logits. Features were standardised with statistics from training windows only. All hyperparameters were fixed before any result was seen and are identical in every condition.

3.6 Evaluation protocol

We used subject-level, stratified, repeated 5-fold cross-validation with five repeats (seeds 0–4), for 25 test splits of 17–18 subjects each. Within each training set, a stratified 20% of subjects formed the early-stopping validation set; it was never used to tune hyperparameters. The fold assignment was generated once, verified by assertions that training, validation and test subjects are pairwise disjoint and that every subject is tested exactly once per repeat, and committed to the code repository before any training run. All conditions use the same 25 splits, so every comparison is paired. The full factorial comprises $7 \times 3 \text{ cells} \times 25 \text{ splits} = 525$ training runs.

The primary metric is subject-level macro-averaged F1. We also report balanced accuracy, one-vs-rest macro ROC-AUC, PR-AUC, per-class sensitivity and specificity, Cohen’s κ , the Brier score, expected calibration error and confusion matrices. Plain accuracy is reported but never used for ranking.

3.7 Baselines, ablations and controls

Baselines. A logistic regression on age and sex alone (the demographic confound floor); a stratified random classifier (chance); and logistic regression, RBF-kernel SVM and random forest trained on the flattened node features of each representation with the same folds, weighting and subject aggregation as the GCN (a non-graph twin of every representation). The MMSE is never used as an input: control participants all score 30, so it is a proxy for the label.

Ablations. Run on the two best cells of the main grid, chosen automatically from the grid results, with the same 25 splits: no graph (identity adjacency, i.e. a per-node MLP); degree-preserving random graphs; $k \in \{3, 6\}$ and a fully connected graph; binarised edges; depth 1 and 3; GAT instead of GCN; no subject-equal weighting; demographic residualisation (age and sex regressed out of every feature, coefficients fitted on training windows); adding P7 (PSWE) features to the best representation; and the PSWE co-occurrence graph (Jaccard overlap of channels’ PSWE time) as the edge type.

Leakage. The best cells were re-run with *window-level* stratified splitting — windows assigned to folds at random, ignoring subject identity — with everything else unchanged.

Permutation test. For the best cell, subject labels were permuted 200 times (all windows of a subject keep one label) and the first repeat’s five folds re-run each time; the p -value is $(1 + \#\{\text{null} \geq \text{observed}\})/(1 + 200)$.

3.8 PSWE detection

Following Milikovsky et al. [22], with parameters as reported in [24], we computed the median power frequency (MPF) of each channel from FFTs of 2 s Hann-windowed segments with 1 s overlap (one MPF value per second) over 1–45 Hz. The frequency range is not reported in the source and was set to our pass-band. A PSWE is a run of at least five consecutive seconds with $\text{MPF} < 6$ Hz. Detection is per channel and per recording, and uses no labels. For each subject we computed the PSWE rate (events per minute, averaged over channels) and the fraction of time spent in PSWEs.

Background-relative PSWE (exploratory). After observing that the fixed-threshold rate was strongly correlated with background slowing (Section 4.8), we added a post-hoc analysis, labelled as such throughout. Its definitions were fixed in the configuration before it was run on any real data. A relative PSWE is a run of at least 5 s in which a channel’s MPF lies below that channel’s own recording-median MPF by at least 2 Hz (primary), 1.5 Hz or 3 Hz, or by at least 2 or 3 robust standard deviations ($1.4826 \times$ median absolute deviation).

3.9 Statistical analysis

The 25 test splits are the repeated-measures units. Main effects of representation and graph construction, and their interaction, were tested with a balanced two-way repeated-measures ANOVA on macro-F1, reporting partial η^2 and Greenhouse–Geisser-corrected p -values. Because cross-validation scores share training data and are therefore not independent, every pairwise comparison is reported with both the Wilcoxon signed-rank test and the Nadeau–Bengio corrected resampled t -test [44] (test/train ratio 17.6/70.4), with Holm correction [45] within each family, Cohen’s d and the matched-pairs rank-biserial correlation. All 21 cells were also compared with a Friedman test

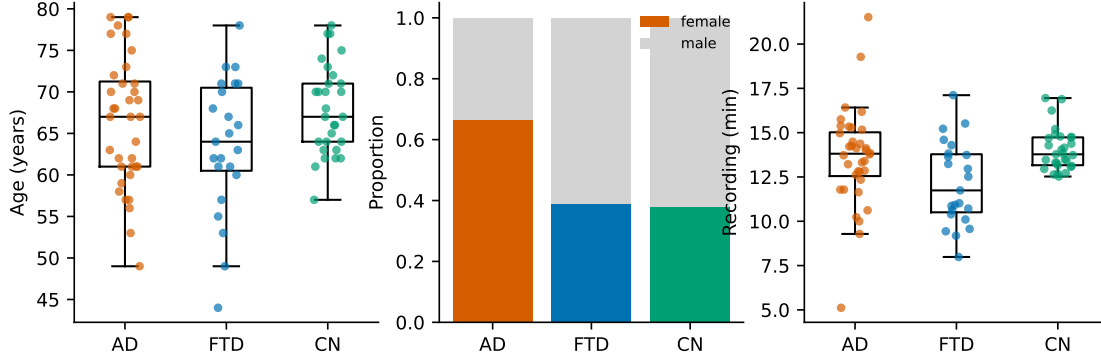


Figure 2: Cohort: age, sex and recording length by group.

and a Nemenyi critical-difference analysis [46]. Cell-level 95% confidence intervals come from a subject-level bootstrap (2,000 resamples of subjects within each repeat) and from a t -interval over the 25 splits.

For PSWEs, group differences were tested with Kruskal–Wallis and Dunn post-hoc tests (Holm-corrected, with rank-biserial effect sizes), and with negative-binomial regression of event counts using recording length as exposure, adjusted for age and sex and, in further models, for relative delta+theta power and background MPF. The relation to slowing was assessed with Spearman and partial (age- and sex-adjusted) rank correlations, and the relation to cognition with Spearman correlation against MMSE within patients only. All analyses were run with SciPy, statsmodels and custom code that is tested against reference implementations.

3.10 Compute

All experiments ran on free Kaggle CPU sessions (4 vCPUs); no GPU was used. The 525-run main grid took 2.4 CPU-hours. Every run is logged with its configuration hash, code version and seed.

4 Results

4.1 Data quality and the confound floor

The fixed preprocessing produced 7,013 windows from all 88 subjects (Figure 2); ICA removed on average 2.4 components per recording. The expected EEG slowing in dementia was present before any model was trained: the median theta/alpha power ratio was 1.57 in AD, 0.68 in FTD and 0.28 in controls, with the posterior alpha of controls replaced by diffuse delta and theta in AD (Figure 3).

A logistic regression on age and sex alone reached a macro-F1 of 0.411, against 0.302 for a stratified random classifier. Demographics therefore carry some group information, consistent with the sex imbalance between groups (Table 2). We use 0.411, not $1/3$, as the floor that EEG-based models must clear.

4.2 Representation matters; graph construction barely does

Across the 21 cells (525 runs; Table 3), the best configuration was $P3 \times \text{spatial}$, with subject-level macro-F1 0.586 ± 0.112 (subject-bootstrap 95% CI 0.538–0.628), balanced accuracy 0.594, macro ROC-AUC 0.749 and Cohen’s κ 0.409.

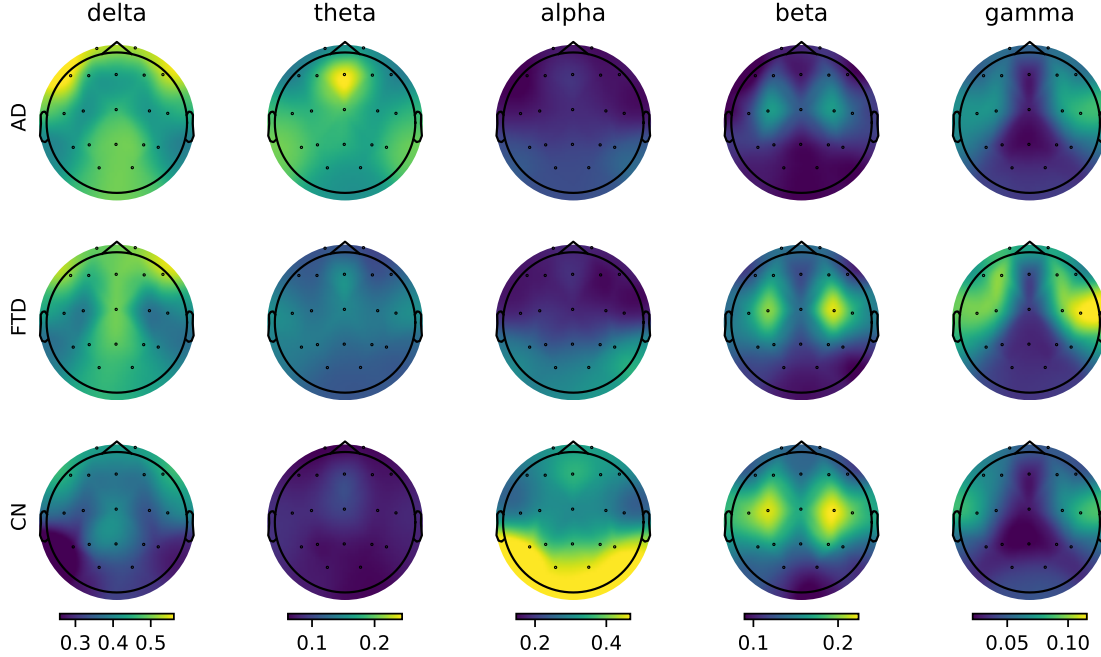


Figure 3: Group-mean relative band power per electrode (rows: groups; columns: bands; colour scales shared within each band).

Representation had a large effect (Table 4; $F(6, 144) = 11.83$, Greenhouse–Geisser $p < 0.001$, partial $\eta^2 = 0.33$). Averaged over graph types, the wavelet (P3, 0.569), spectral (P1, 0.552) and fused (P6, 0.543) representations formed a leading group; raw windows (P5, 0.489) were intermediate; and connectivity strength (P2, 0.453), PSWE features (P7, 0.448) and time-domain statistics (P4, 0.434) trailed (Figure 4). Graph construction had a small effect ($F(2, 48) = 4.03$, $p = 0.026$, partial $\eta^2 = 0.14$): marginal macro-F1 was 0.506 for spatial, 0.497 for functional and 0.492 for hybrid graphs (Figure 4). The representation $\times$ graph interaction was not significant ($F(12, 288) = 1.43$, $p = 0.194$, partial $\eta^2 = 0.06$; Figure 5). A Friedman test across all cells was significant ($p < 0.001$), but the Nemenyi critical difference (6.26 ranks) left the top cells statistically indistinguishable from one another (Figure 6).

Pairwise comparisons depend on how the dependence between cross-validation splits is handled. With Holm-corrected Wilcoxon tests, 9 of the 21 representation pairs differed; with the more conservative Nadeau–Bengio corrected t -test, 0 did. For example, P3 exceeded P4 by 0.135 macro-F1 (Cohen’s $d = 1.50$; Wilcoxon–Holm $p < 0.001$; Nadeau–Bengio–Holm $p = 0.216$). We regard the omnibus effect and the consistent ordering as robust, and individual pairwise differences as suggestive.

4.3 What the models get right and wrong

For P3 $\times$ spatial, sensitivity was 0.771 for controls, 0.642 for AD and only 0.368 for FTD (specificities 0.786, 0.775 and 0.852). Across representations, controls were recognised best and FTD — the smallest group — worst; FTD recordings were most often misclassified as AD, except with PSWE features, where they were more often taken for controls (Figure 7). The ROC and precision–recall curves show the same ordering (Figure 8). The models were poorly calibrated (expected calibration error 0.265), so their probabilities should not be read as clinical risks.

Table 3: Subject-level performance of every cell (mean $\pm$ SD over 25 test splits). CI: subject-bootstrap 95% interval of macro-F1. BA: balanced accuracy; AUC: one-vs-rest macro ROC-AUC.

Representation	Graph	Macro-F1	95% CI	BA	AUC	κ	Brier
P1 spectral	functional	0.551 $\pm$ 0.100	[0.507, 0.603]	0.565	0.724	0.363	0.578
P1 spectral	hybrid	0.545 $\pm$ 0.117	[0.504, 0.600]	0.561	0.717	0.357	0.599
P1 spectral	spatial	0.560 $\pm$ 0.103	[0.519, 0.616]	0.575	0.720	0.378	0.593
P2 connectivity	functional	0.455 $\pm$ 0.108	[0.418, 0.507]	0.477	0.660	0.215	0.626
P2 connectivity	hybrid	0.451 $\pm$ 0.090	[0.419, 0.507]	0.479	0.659	0.222	0.628
P2 connectivity	spatial	0.454 $\pm$ 0.102	[0.420, 0.509]	0.483	0.653	0.225	0.629
P3 wavelet	functional	0.553 $\pm$ 0.106	[0.503, 0.597]	0.560	0.724	0.359	0.598
P3 wavelet	hybrid	0.568 $\pm$ 0.103	[0.522, 0.614]	0.578	0.735	0.384	0.582
P3 wavelet	spatial	0.586 $\pm$ 0.112	[0.538, 0.628]	0.594	0.749	0.409	0.558
P4 time domain	functional	0.418 $\pm$ 0.103	[0.382, 0.473]	0.436	0.623	0.163	0.668
P4 time domain	hybrid	0.430 $\pm$ 0.104	[0.394, 0.488]	0.450	0.608	0.182	0.677
P4 time domain	spatial	0.456 $\pm$ 0.099	[0.418, 0.511]	0.474	0.611	0.217	0.677
P5 raw	functional	0.504 $\pm$ 0.087	[0.469, 0.556]	0.528	0.697	0.320	0.596
P5 raw	hybrid	0.470 $\pm$ 0.086	[0.435, 0.514]	0.496	0.687	0.275	0.600
P5 raw	spatial	0.494 $\pm$ 0.097	[0.458, 0.544]	0.526	0.706	0.318	0.594
P6 fused	functional	0.556 $\pm$ 0.099	[0.506, 0.599]	0.571	0.731	0.370	0.605
P6 fused	hybrid	0.532 $\pm$ 0.099	[0.490, 0.585]	0.554	0.724	0.342	0.617
P6 fused	spatial	0.540 $\pm$ 0.094	[0.496, 0.589]	0.558	0.738	0.349	0.589
P7 PSWE	functional	0.444 $\pm$ 0.085	[0.407, 0.498]	0.467	0.666	0.220	0.615
P7 PSWE	hybrid	0.445 $\pm$ 0.084	[0.404, 0.495]	0.466	0.670	0.218	0.613
P7 PSWE	spatial	0.454 $\pm$ 0.082	[0.415, 0.504]	0.472	0.662	0.225	0.620

Table 4: Two-way repeated-measures ANOVA on subject-level macro-F1 (units: 25 test splits). ε : Greenhouse–Geisser; p : GG-corrected.

Effect	df	F	ε	p	partial η^2
Representation	6, 144	11.83	0.71	<0.001	0.33
Graph construction	2, 48	4.03	0.96	0.026	0.14
Interaction	12, 288	1.43	0.60	0.194	0.06

4.4 Graph models do not beat their non-graph twins

Classical classifiers trained on the same features, folds and weighting reached the same ceiling as the GCN (Table 5). The best non-graph model on spectral features (a random forest) scored 0.588, and on wavelet features 0.581. The paired difference between each representation’s best GCN cell and its best non-graph twin ranged from -0.048 (P7) to $+0.020$ (P2), and none was significant after correction (for P3: $+0.005$, $p = 1.000$). The spectral, wavelet, fused and raw representations cleared the demographic floor of 0.411; the connectivity (P2), time-domain (P4) and PSWE (P7) representations scored within about 0.05 of it.

4.5 Ablations

Table 6 and Figure 9 report each ablation as a paired difference from its reference cell. No ablation reached significance under the Nadeau–Bengio test with Holm correction. Removing the graph altogether (identity adjacency) lowered macro-F1 by -0.043 [-0.088 , $+0.003$] for the spatial cell and by -0.024 for the hybrid cell. A degree-preserving random graph cost about the same (-0.037 and -0.018). Neither difference is significant, but the pattern is consistent with any benefit of message passing coming from smoothing over neighbours rather than from a specific electrode topology. A fully connected graph was the only structural change whose t -interval excluded zero for the spatial cell (-0.066 [-0.113 , -0.020]), consistent with over-smoothing. Sparsity ($k = 3$ or 6), binarised

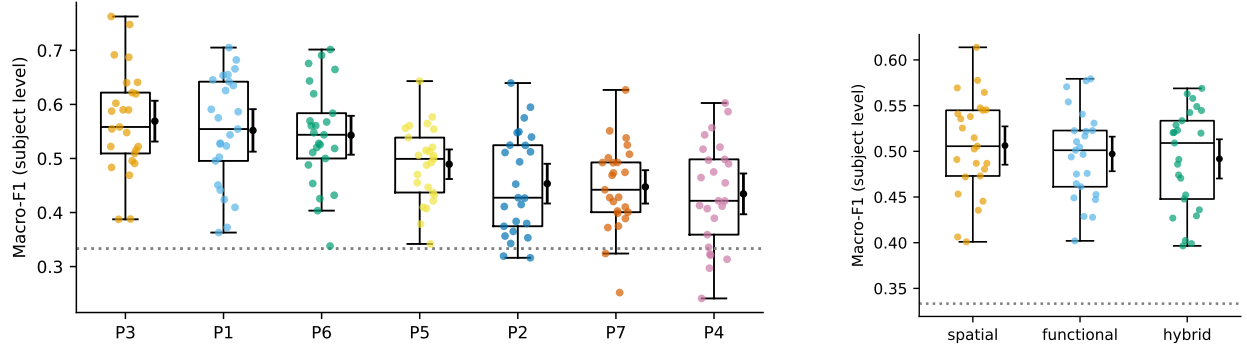


Figure 4: Subject-level macro-F1 by representation (left, averaged over graph types) and by graph construction (right, averaged over representations). Points are the 25 test splits; black markers, mean $\pm$ 95% CI; dotted line, $1/3$.

edges, depth, GAT attention and loss weighting all changed macro-F1 by less than 0.03. Regressing age and sex out of every feature cost -0.046 $[-0.089, -0.003]$, leaving performance well above the demographic floor.

4.6 Leakage inflates results to the published range

Re-running the two best cells with window-level splitting — identical data, features, graph, model and hyperparameters — raised subject-level macro-F1 from 0.586 to 0.988 (spatial) and from 0.568 to 0.974 (hybrid). Window-level macro-F1, the metric most often reported, rose to 0.842 and 0.830 (Figure 10). The leaky protocol thus reproduces the 95–99% figures reported on this benchmark with a model that, evaluated correctly, scores below 0.6.

4.7 The best cell is above chance

Under 200 label permutations, the null macro-F1 of $P3 \times \text{spatial}$ averaged 0.313 (maximum 0.474), against an observed 0.575 on the same splits ($p = 0.005$, the smallest value attainable with 200 permutations; Figure S2).

4.8 PSWE: a BBB-associated marker that tracks slowing

Fixed-threshold PSWEs were most frequent in AD (median 1.34 events/min), intermediate in FTD (0.41) and least frequent in controls (0.22; Kruskal–Wallis $p = 0.041$; Figures 11 and 12). In negative-binomial models adjusted for age and sex, the PSWE rate ratio relative to controls was 2.96 (95% CI 1.58–5.57, $p < 0.001$) in AD and 2.07 (1.03–4.15, $p = 0.041$) in FTD.

The PSWE rate was, however, almost collinear with background slowing (Spearman $\rho = 0.88$ with relative delta+theta power; age- and sex-adjusted partial $\rho = 0.87$). Once slowing was added to the model, the AD rate ratio reversed to 0.50 (0.33–0.75, $p < 0.001$), and FTD fell to 0.67 ($p = 0.051$). The PSWE rate was not significantly associated with MMSE within patients ($\rho = -0.22$, $p = 0.095$). In the classifier, PSWE features alone (P7) performed near the demographic floor (0.448), and adding them to the best representation, or using PSWE co-occurrence as the graph, did not help (Section 4.5).

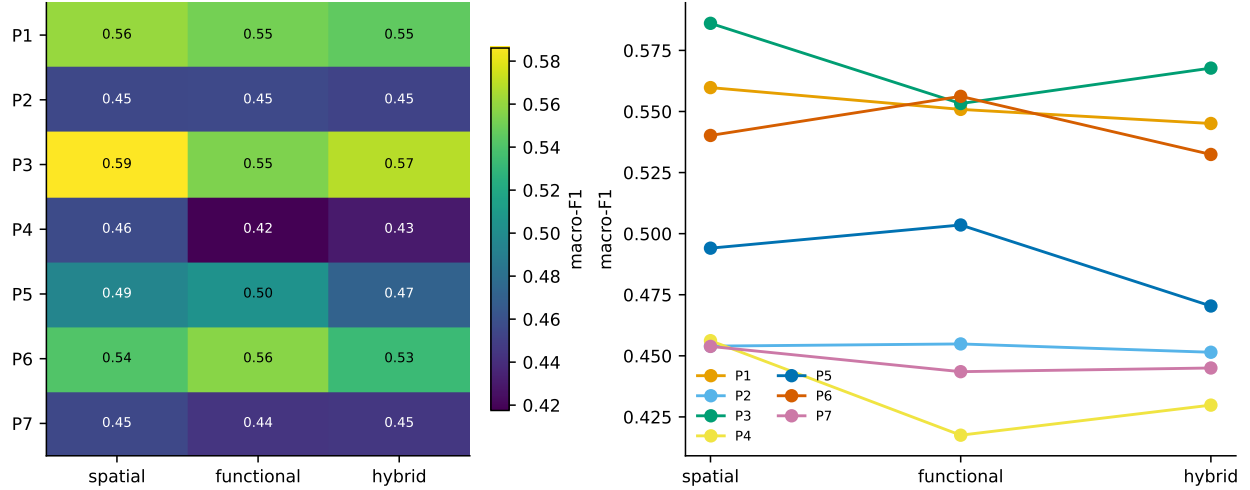


Figure 5: Mean macro-F1 of every representation $\times$ graph cell (left) and the same values as an interaction plot (right). Lines are close to parallel: the best representation does not depend on how the graph is built.

Background-relative PSWEs (exploratory). Defining PSWEs relative to each channel’s own background reversed the group pattern. With the primary definition (a drop of at least 2 Hz below the channel median), AD showed *fewer* transient slowing events than controls (rate ratio 0.55, 0.35–0.89, $p = 0.014$), FTD a non-significant reduction (0.70), and the event rate was only weakly related to slowing ($\rho = -0.15$) and not to MMSE ($\rho = 0.04$). After additional adjustment for background MPF the AD effect was no longer significant (0.77, $p = 0.299$). The MAD-scaled variants gave the same direction more strongly and survived that adjustment (for 2 MAD: AD 0.43, $p = 0.014$; FTD 0.35, $p = 0.002$). Every variant pointed the same way: relative to their own background, patients with dementia had fewer, not more, transient slowing events.

5 Discussion

5.1 Principal findings

Holding the model, data and evaluation protocol fixed, we found four things. First, the node-feature representation strongly affects three-class AD/FTD/control classification (partial $\eta^2 = 0.33$), with wavelet, spectral and fused features leading. Second, how the graph is built matters little (partial $\eta^2 = 0.14$), the best representation does not depend on it, and the GCN performs no better than a random forest or SVM on the same features. Third, splitting windows instead of subjects inflates subject-level macro-F1 from about 0.59 to about 0.99 on the identical pipeline, reproducing the top of the published range. Fourth, the blood–brain-barrier-associated PSWE marker is more frequent in AD and FTD than in controls, but this excess is explained by continuous background slowing; defined relative to each person’s background, transient slowing events are, if anything, less frequent in dementia.

5.2 Representation matters more than topology

The size and consistency of the representation effect, against a small and non-interacting graph effect, indicate that the information a GNN can use on this benchmark lives mostly in the node features.

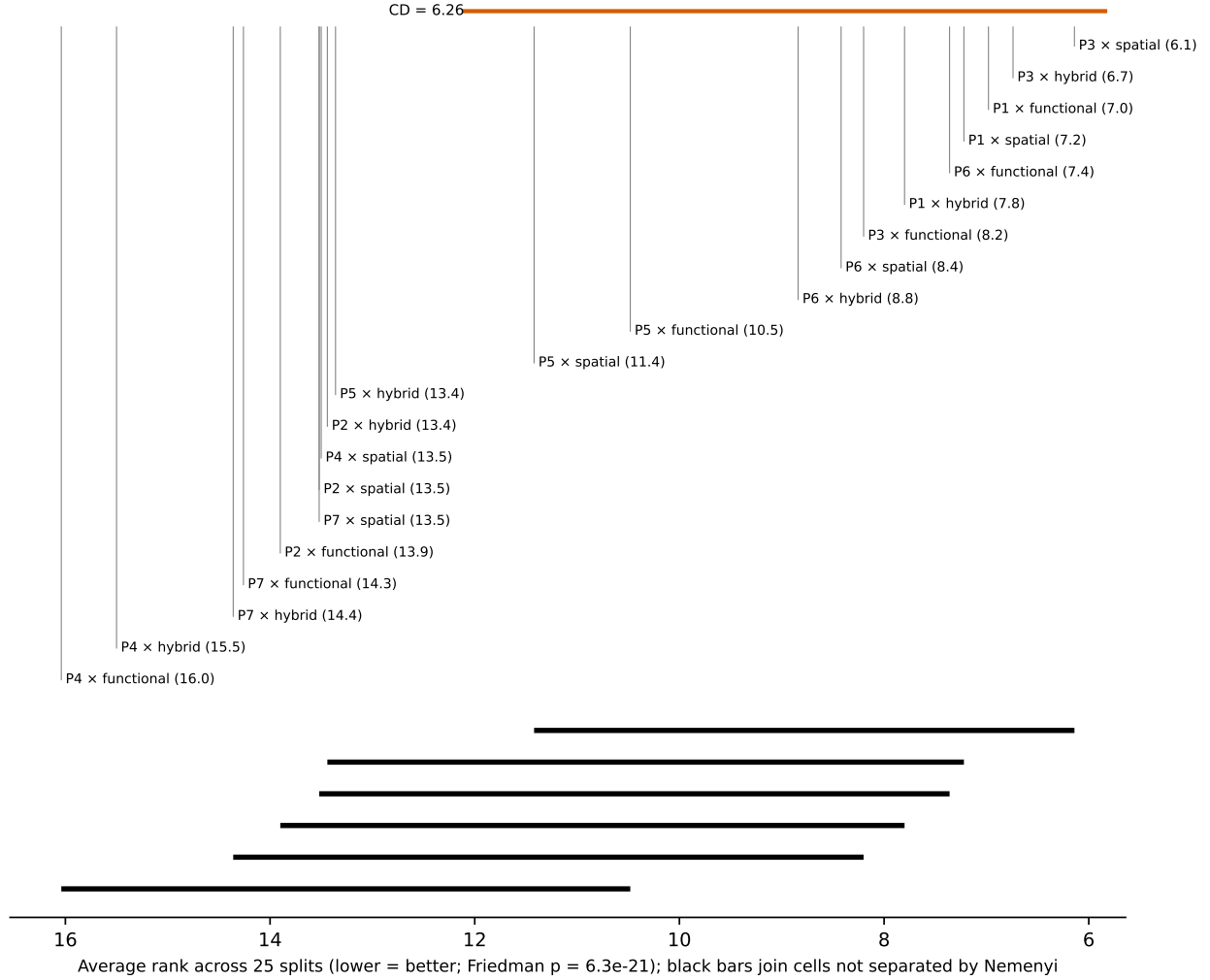


Figure 6: Critical-difference diagram over the 21 cells. Cells joined by a black bar are not separated by the Nemenyi test.

Several observations point the same way. Spatial graphs — which are identical for every subject and therefore carry no diagnostic information of their own — did at least as well as functional and hybrid graphs. A degree-preserving random graph cost about as much as removing the graph altogether, which suggests that message passing helps, where it helps at all, by smoothing features over neighbouring electrodes rather than by exploiting a particular connectivity pattern. And non-graph classifiers on flattened features matched the GCN cell for cell. These results agree with the independent conclusion of the 2026 scoping review that apparent representational gains on this dataset do not survive leakage-free evaluation [15], and they temper the case for graph-based modelling made in recent work [17–19]: on 19-electrode scalp EEG from 88 subjects, we see no evidence that graph structure adds information beyond good node features. Whether this holds with source-space nodes, learned adjacency or much larger cohorts remains open.

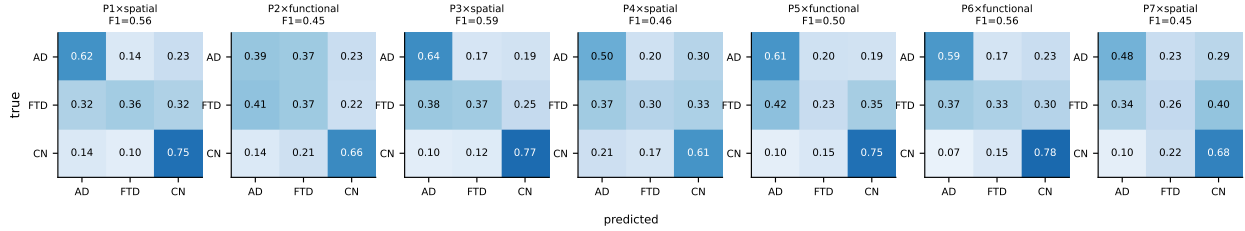


Figure 7: Row-normalised confusion matrices summed over the 25 test splits, for the best graph type of each representation.

5.3 Neurophysiological interpretation

Representations that describe each electrode’s own spectrum (P1, P3) outperformed one built only from inter-electrode phase coupling (P2), and the functional graph did not improve on the spatial one. Within the limits of scalp EEG, this favours local oscillatory slowing over altered large-scale coupling as the dominant discriminative signal, consistent with spectral slowing being the most replicated EEG marker of AD [27, 28] and with the posterior-alpha-to-diffuse-theta shift visible in our data (Figure 3). It does not contradict network accounts of dementia [1, 2]: alpha-band wPLI estimated within 10s windows is noisy, and electrode-level graphs are a coarse proxy for cortical networks. The error pattern is clinically recognisable: controls are identified best, while FTD — the smallest group, and less slowed than AD — is identified worst and is usually mistaken for AD. Separating the two dementias remains the hard part of the problem.

5.4 What the PSWE results mean for an EEG marker of BBB dysfunction

With the fixed threshold, our data agree in direction with Milikovsky et al.: PSWE rates were roughly three times higher in AD than in controls after adjusting for age and sex, and FTD — not previously studied — was also elevated. Two further analyses change the interpretation. The fixed-threshold rate was almost collinear with relative delta+theta power, and adjusting for that slowing reversed the AD effect. When events were instead defined as transient drops below each channel’s own background, AD and FTD showed fewer of them than controls, under every variant we examined. Together these results suggest that, in resting scalp EEG with diffuse slowing, a median power frequency below 6 Hz mostly flags a slow *background* rather than *paroxysmal* events.

We draw three cautious conclusions. First, a fixed-threshold PSWE rate should not be read as evidence of BBB dysfunction without controlling for background slowing. Second, background-relative definitions are needed if PSWEs are to capture transient cortical dysfunction, although they bring their own floor effect in very slow recordings, which we observed in the Hz-drop variants. Third, our data cannot test the BBB link itself: ds004504 contains no BBB imaging, the original detection details include parameters we could not verify from the full text, and Milikovsky et al. studied different patients under different recording conditions. The decisive test is a cohort with both EEG and dynamic contrast-enhanced MRI.

5.5 Implications for how results on this benchmark are read

On one pipeline, the choice between subject-level and window-level splitting moved subject-level macro-F1 by about 0.4 — roughly three times the gap between the best and worst representation, and several times larger than any graph or architecture change we tested. Figures in the high 90s on

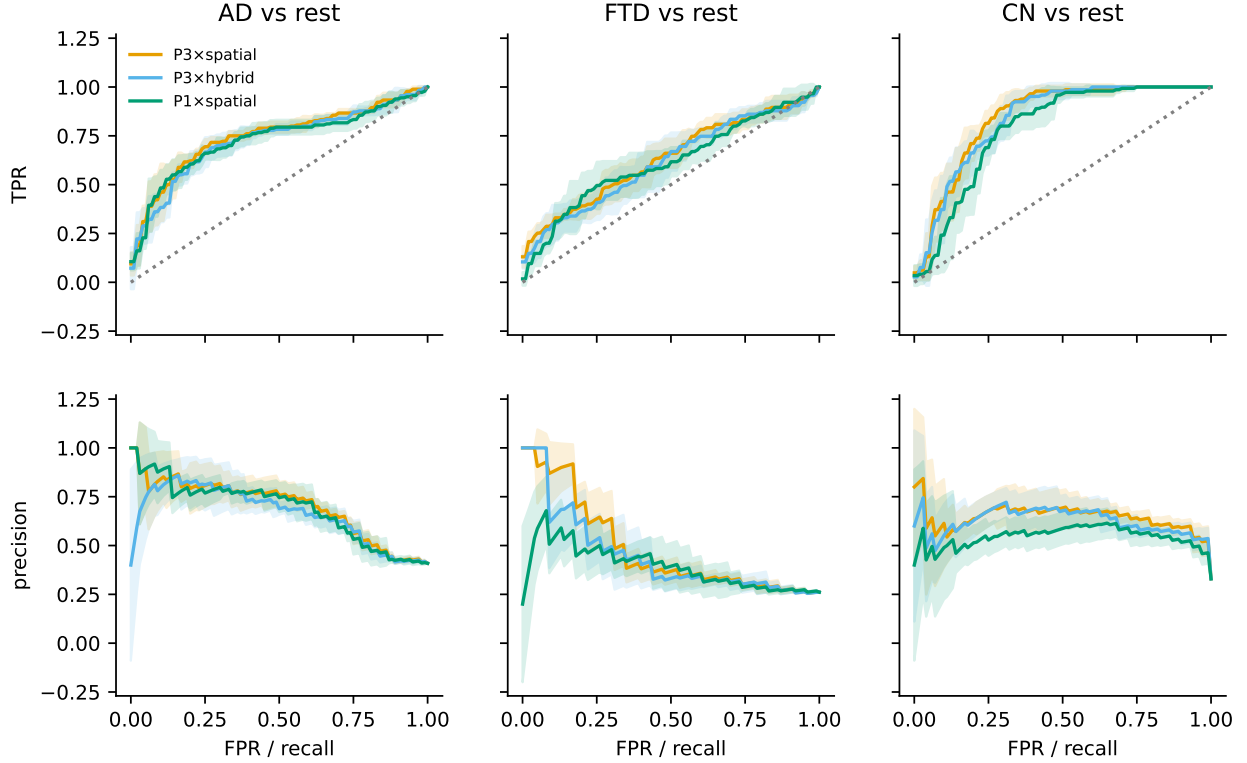


Figure 8: One-vs-rest ROC (top) and precision–recall (bottom) curves for the three best cells; mean $\pm$ SD across the five repeats.

ds004504 should therefore be read alongside their splitting protocol, and studies that report window-level metrics or allow a subject’s windows into both training and test sets cannot be compared with subject-level results. We name no individual study: the point is that the protocol, not the model, determines most of the reported number, as argued independently in [15, 16]. The permutation test confirms that the honest result is nonetheless real.

5.6 Limitations

The cohort is small: test folds contain 17–18 subjects, and under the conservative Nadeau–Bengio correction no pairwise difference between representations, and no single ablation, reached significance. We report both corrected and uncorrected tests and give effect sizes, but small differences between leading representations should not be over-interpreted. All data come from one site, device and protocol, and FTD is the smallest class ($n = 23$). The planned external validation on an independent cohort (ds004584) and two planned ablations (window length and ICA on/off) were not run for this draft. Hyperparameters were fixed a priori and not tuned; early stopping on a validation set of about 14 subjects stopped training early, which may disadvantage higher-capacity configurations such as raw windows (P5, about 84k parameters). Calibration was poor. Graphs are defined on electrodes rather than cortical sources. The demographic floor (0.411) shows residual age and sex information, which residualisation reduced but did not remove. For PSWEs, the MPF frequency range was not reported in the source and was set by us, we detected events per channel rather than over averaged regions, the background-relative analysis is post hoc, and no BBB measure is available.

Table 5: Best GCN cell of each representation against its non-graph twins (same folds, weighting and subject aggregation) and the demographic floor. Macro-F1, mean $\pm$ SD over 25 splits.

Representation	Model	Macro-F1
P1 spectral	Random forest	0.588 ± 0.117
P1 spectral	Logistic regression	0.564 ± 0.096
P1 spectral	RBF-SVM	0.560 ± 0.108
P1 spectral	GCN (spatial)	0.560 ± 0.103
P2 connectivity	GCN (functional)	0.455 ± 0.108
P2 connectivity	RBF-SVM	0.435 ± 0.089
P2 connectivity	Random forest	0.418 ± 0.055
P2 connectivity	Logistic regression	0.417 ± 0.082
P3 wavelet	GCN (spatial)	0.586 ± 0.112
P3 wavelet	Random forest	0.581 ± 0.092
P3 wavelet	RBF-SVM	0.543 ± 0.122
P3 wavelet	Logistic regression	0.466 ± 0.126
P4 time domain	Random forest	0.487 ± 0.123
P4 time domain	RBF-SVM	0.483 ± 0.103
P4 time domain	GCN (spatial)	0.456 ± 0.099
P4 time domain	Logistic regression	0.446 ± 0.112
P5 raw	GCN (functional)	0.504 ± 0.087
P6 fused	RBF-SVM	0.579 ± 0.098
P6 fused	Random forest	0.578 ± 0.118
P6 fused	GCN (functional)	0.556 ± 0.099
P6 fused	Logistic regression	0.514 ± 0.118
P7 PSWE	Random forest	0.502 ± 0.077
P7 PSWE	RBF-SVM	0.473 ± 0.088
P7 PSWE	Logistic regression	0.456 ± 0.066
P7 PSWE	GCN (spatial)	0.454 ± 0.082
Demographics only	Logistic regression (age + sex)	0.411 ± 0.120
Demographics only	Stratified chance	0.302 ± 0.110

5.7 Future work

The most valuable next study is a multi-site replication of this exact protocol, since a single-site design cannot establish device or population invariance. Natural extensions are source-space graphs [19], learned adjacency [8], spatio-temporal graphs over window sequences [9, 11], an architecture sweep under a fixed representation (the transpose of this design) [6], an MCI class, and — for the BBB question — simultaneous EEG and dynamic contrast-enhanced MRI to test whether any PSWE definition tracks measured barrier leakage [20, 21].

6 Conclusion

We asked whether the performance of graph neural networks for EEG-based dementia classification comes from how the signal is represented, how the brain graph is built, or the model. On a public 88-subject benchmark, with the model, data and evaluation protocol held fixed and seven representations crossed with three graph constructions over 25 paired subject-level splits, the answer

Table 6: Ablations: paired Δ macro-F1 (ablated minus reference) with 95% t -interval over the 25 splits, and Nadeau–Bengio p with Holm correction within each cell.

Ablation	P3 $\times$ spatial		P3 $\times$ hybrid	
	Δ [95% CI]	p_{Holm}	Δ [95% CI]	p_{Holm}
Age/sex regressed out	-0.046 [-0.089, -0.003]	1.00	-0.044 [-0.093, +0.004]	1.00
No graph (identity)	-0.043 [-0.088, +0.003]	1.00	-0.024 [-0.064, +0.015]	1.00
Random graph (degree-preserving)	-0.037 [-0.075, +0.002]	1.00	-0.018 [-0.049, +0.013]	1.00
Fully connected	-0.066 [-0.113, -0.020]	1.00	-0.032 [-0.068, +0.005]	1.00
$k = 3$	-0.022 [-0.047, +0.004]	1.00	-0.009 [-0.026, +0.007]	1.00
$k = 6$	-0.009 [-0.033, +0.015]	1.00	+0.000 [-0.031, +0.032]	1.00
Binarised edges	-0.006 [-0.020, +0.008]	1.00	-0.023 [-0.057, +0.011]	1.00
Depth 1	-0.026 [-0.073, +0.020]	1.00	-0.006 [-0.042, +0.031]	1.00
Depth 3	-0.010 [-0.042, +0.022]	1.00	+0.007 [-0.018, +0.033]	1.00
GAT instead of GCN	-0.022 [-0.069, +0.025]	1.00	-0.021 [-0.044, +0.002]	1.00
No subject-equal weighting	+0.007 [-0.030, +0.044]	1.00	+0.014 [-0.016, +0.044]	1.00
+ P7 PSWE features	-0.024 [-0.059, +0.011]	1.00	+0.005 [-0.028, +0.038]	1.00
PSWE co-occurrence edges	-0.029 [-0.065, +0.007]	1.00	-0.011 [-0.044, +0.022]	1.00

is mostly the representation: wavelet and spectral node features lead, graph construction matters little, and the GCN does not outperform non-graph classifiers on the same features. The best honest three-class macro-F1 was 0.586, clearly above chance ($p = 0.005$) but far below published figures — a gap that window-level splitting alone closes on our own pipeline, inflating subject-level macro-F1 to 0.988. A blood–brain-barrier-associated EEG marker, the paroxysmal slow-wave event, was elevated in AD and FTD under its original definition but was explained by background slowing, and defining it relative to each person’s background reversed the effect. We release the code, seeds, frozen folds and every raw run so that each number in this paper can be regenerated from the data.

Data and code availability

The EEG data are OpenNeuro ds004504 (CC0) [14]. Code, configuration (`configs/default.yaml`), the frozen fold assignment (`splits/folds_ds004504.json`, with the SHA-256 of `participants.tsv`), every raw training run (one JSON record per run, with configuration hash, code version, seed and per-subject test predictions), and the Kaggle notebooks that produced them are available at https://github.com/DevGuru-codes/Major_project_IIITN. Random seeds are 0–4 for the main grid and ablations, and $10,000 + i$ for permutation i . Every table and figure in this paper is regenerated from the committed raw runs by a single command (`python -m eegrep report`), and the resolved package versions are recorded in `reports/kaggle_pip_freeze.txt`. The full study runs on free cloud CPUs.

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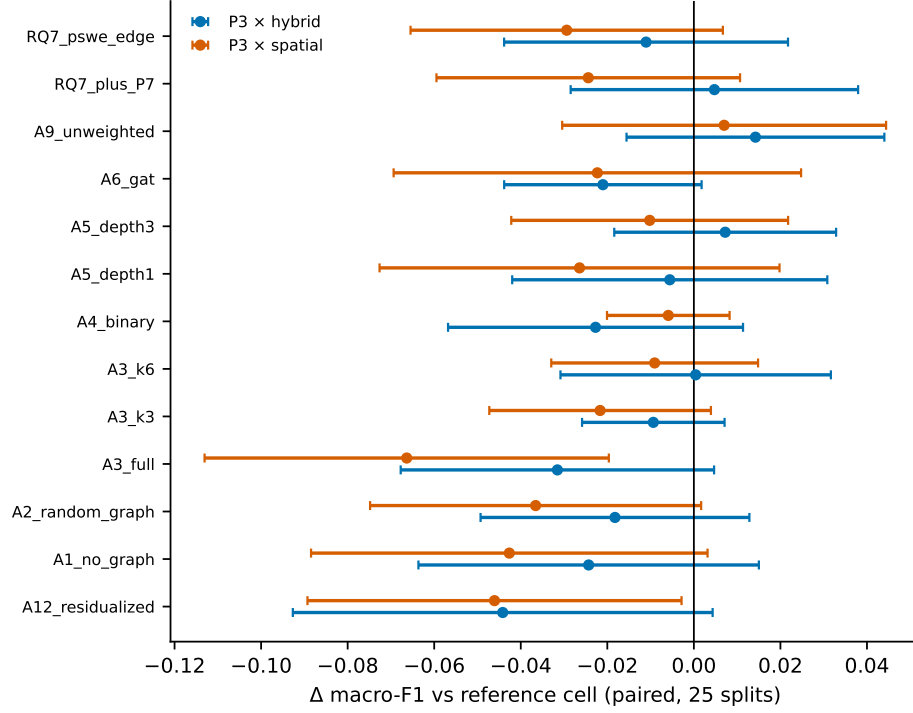


Figure 9: Ablations as paired differences from the two reference cells (mean and 95% CI over 25 splits). An asterisk would mark Nadeau–Bengio–Holm $p < 0.05$; none qualifies.

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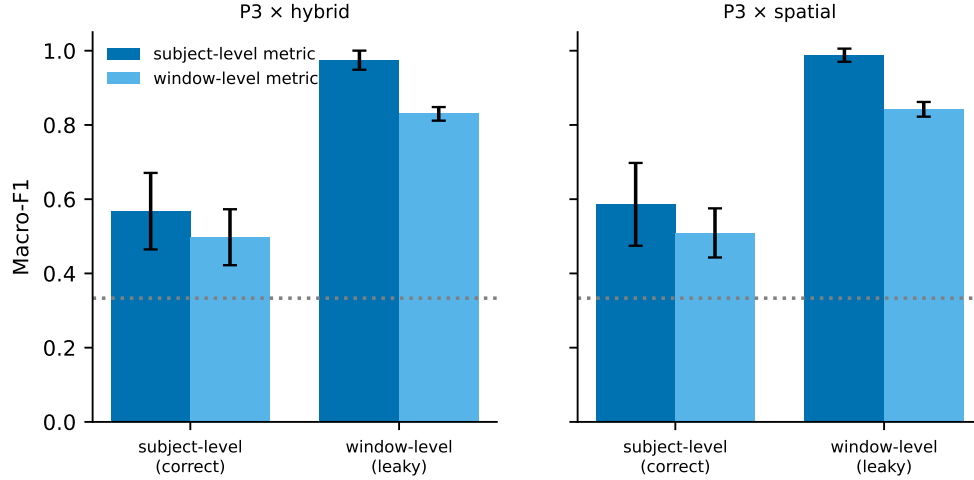


Figure 10: The same pipeline evaluated with subject-level (correct) and window-level (leaky) splitting. Bars: mean $\pm$ SD over 25 splits; dotted line, 1/3.

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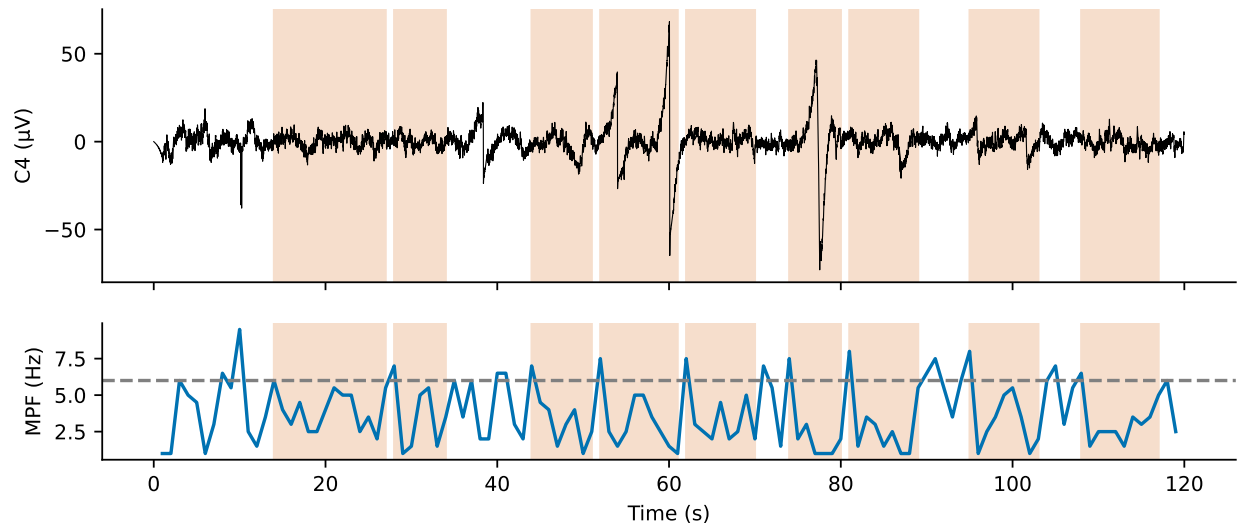


Figure 11: Fixed-threshold PSWE detection in the AD recording with the highest PSWE rate among the example shard (first 120s, channel with most events). Top: EEG; bottom: median power frequency with the 6 Hz threshold; shaded, detected PSWEs. Because this subject’s background is already slow, most of the recording is flagged — the pattern behind the collinearity with slowing.

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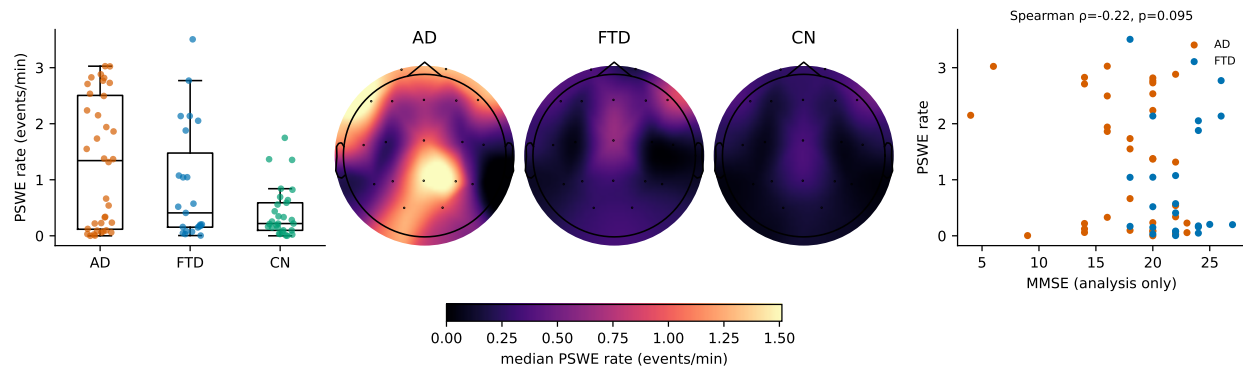


Figure 12: Fixed-threshold PSWE rate per subject by group (left), median per-channel rate as scalp maps (middle), and rate against MMSE in patients (right; analysis only).

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Supplementary material

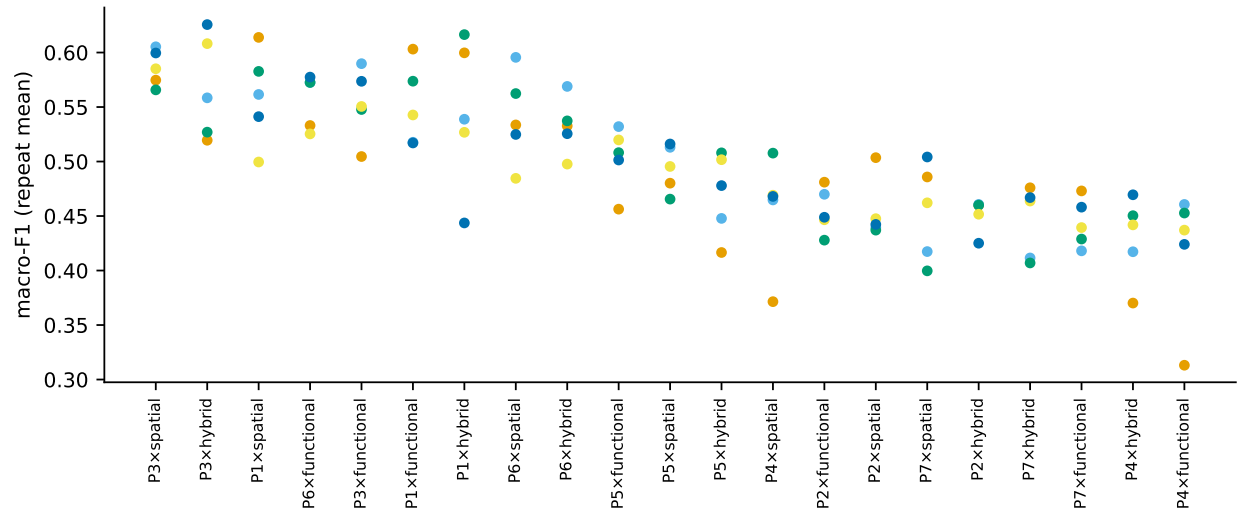


Figure S1: Mean macro-F1 of each repeat (seed) for every cell: variation due to the cross-validation partition and initialisation.

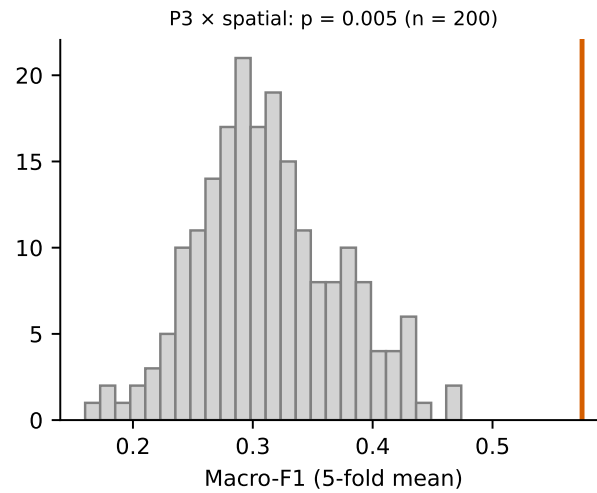


Figure S2: Label-permutation null distribution (200 permutations, first repeat's five folds) for the best cell; red line, observed macro-F1 on the same splits.